

Review

AI integration in construction safety: Current state, challenges, and future opportunities in text, vision, and audio based applications

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ABSTRACT

High occupational injury and fatality rate in the construction industry is a serious global concern. Recognizing AI as a solution to enhance safety performance, this study reviews 153 papers to assess and categorize current AI applications in construction, focusing on text, visual, and audio data, while also identifying challenges and future research opportunities. Real-time monitoring, hazard detection, and information extraction are identified as key areas where AI is applied, with a notable reliance on deep neural networks, object recognition, and Natural Language Processing. The review highlights major challenges, including the need for high-quality data management, semantic feature representation, and occluded object detection. Additionally, it underscores the untapped potential of audio-based AI and the advancements possible with Large Language Models for text interpretation. The findings emphasize the need for integrated, multi-faceted AI systems and advocate for responsible AI deployment to mitigate safety risks on construction sites.

1. Introduction

Since the dawn of modern civilization construction has always been considered one of the most important yet most dangerous industries in the world. The persistent issue of occupational injuries and fatalities on a global scale is underscored by the grim frequency of one death approximately every nine minutes [1]. In the US, one in every five workplace deaths is reported from the construction industry while employing just over 6% of the workforce [2]. In 2021, the US Bureau of Labor Statistics reported 1015 deaths in construction. Almost 40% of these were due to falls, slips, and trips, marking a 7.2% increase from 2020 [3]. Similarly, China's construction domain remains one of its most dangerous industries, with fatality rates showing no significant decline over last few decades [4]. European statistics on the subject are equally bleak, reporting more than 700 fatal construction incidents annually, equating to a fatality incidence of 13 per 100,000 workers [5]. Complementing these findings, research by the Center for Construction Research and Training (CPWR) demonstrates that a construction professional, over a span of 45 years in the industry, faces a 75% likelihood of enduring serious injuries [6]. Beyond the incalculable human toll, the economic ramifications of these disproportionate injury rates are significant. The fiscal burden of occupational injuries in the construction

industry can reach up to 3% of total project costs, equating to 10% of labor expenses in the U.S. [7].

The construction industry, inherently characterized by its complex and dynamic nature, generates substantial volumes of data across various processes. The data produced in construction processes predominantly manifests in text, visual, and audio formats [8]. Leveraging AI techniques such as data mining, Natural Language Processing (NLP), object and action recognition, and semantic segmentation presents a strategic approach for deriving valuable safety insights from the extensive datasets originating from construction sites. With advancements in Artificial Intelligence (AI), the construction sector is exploring the use of abundant digital data for transformation of safety management processes. AI algorithms allow for real-time monitoring of construction environments and personnel, aiding in the early identification of potential hazards and prompting preventive measures. Moreover, through efficient analysis of existing data (such as past accident reports), AI can discern patterns that inform future safety protocols. For instance, insights derived from this analysis (such as frequent accident types, contributing factors) can lead to enhanced worker training or modifications in existing work procedures [9]. AI-empowered drones and robots can also efficiently inspect construction sites, reducing human risks [10] and increasing operational efficiency.

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In addition to the research interest in the topic, there is a growing interest in the construction industry in adopting AI technologies for various management and construction tasks [11]. These include the use of computer vision for real-time safety monitoring, quality control, and progress monitoring [12,13]. Similarly, there is a growing interest in leveraging NLP and Large Language Models for safety management, automating tasks like safety reporting and risk evaluation [14]. In addition to software solutions, the industry is also integrating robotics for various construction tasks, with robots like MULE [15], dusty [16], SAM [17] gaining popularity.

1.1. Goals and study focus

The purpose of this study is to examine the current state of AI usage in improving construction safety, identify the challenges that impede its adoption, and opportunities for future research. This investigation focuses on applications of AI that utilize text, visual, and audio data. Firstly, these are the most prevalent forms of data generated in construction. Text data (such as reports, logs, and communications) can provide detailed accounts of incidents and operational procedures, visual data can capture real-time conditions and activities on-site for visual inspection and hazard identification, and audio data can offer insights into ambient conditions, machinery operations. Secondly, there has been rapid advancement in AI capabilities specific to these data types which offers unique opportunities for safety applications. Finally, these data types, closely mimic human sensory inputs (sight, hearing, and reading/comprehension), allowing AI systems to emulate human perception in assessing safety conditions. Hence, this focus allowed for a comprehensive review while maintaining a distinct categorization of literature.

Furthermore, while several advanced technologies are employed in construction safety, this review specifically focuses on literature that employs AI approaches (such as machine learning techniques, neural networks, natural language processing and computer vision algorithms) to emulate human cognitive functions such as perception, learning, problem-solving, and decision-making. Additionally, while AI applications in construction safety may use various data modalities, our review is deliberately limited to text, visual, and audio data. This limitation is intended to provide a focused examination within these prevalent modalities.

1.2. Point of departure

As of 2023, there have been several review papers examining different AI applications in construction safety. However, two clear gaps are evident-1) The majority of existing reviews on AI applications in the construction industry tend to focus on specific domains within AI, such as computer vision [18–21], NLP [22–24] or information technology [25,26], thereby addressing only fragments of the potential AI applications. While these studies are valuable, they may not provide a complete picture of how various AI techniques and data sources can collaboratively contribute to construction safety. In contrast, our study distinguishes itself by providing an updated review of AI applications that integrates text, visual, and audio data-based AI applications. This approach allows a thorough understanding of individual AI techniques with respect to each data modalities and examines their corresponding challenges and future research possibilities ensuring a more detail exploration of AI-assisted safety management possibilities. 2) While there are some review studies that take a broad perspective on AI applications in construction, they often encompass a wide range of applications such as quality inspections, productivity assessment, and more [27–29]. In contrast, our review specifically focuses on construction safety. This targeted approach ensures a detailed and in-depth exploration of safety-related AI applications that can potentially get overshadowed in broader reviews.

2. Background

The following section outlines a brief overview of construction safety management and Artificial Intelligence, a prerequisite to better understand the applications of AI in construction safety management.

2.1. Construction safety management: a brief overview

Construction Safety Management (CSM) is a nexus of strategies, practices, and methodologies designed explicitly to safeguard construction workforce. At its core, CSM seeks to systematically anticipate, identify, and prevent risks that could harm construction workers, the broader public, or even the infrastructure itself [30]. The transient nature of construction sites, alongside with the constant interplay of man, machine, and materials, frequently exposes them susceptible to unforeseen hazards. Frequent incidents, such as ‘falls from heights’ or ‘struck by hazards’ highlight not just their regular occurrences but possible systemic challenges in safety protocols [31]. These incidents, while offering lessons to enhance safety, also show potential shortcomings in existing measures. Despite the advancements in AI and its potential to revolutionize safety measures, the construction industry remains somewhat hesitant to fully embrace this digital evolution. This reluctance stems from factors like the diverse nature of projects, financial limitations, and range of worker expertise. The era of Artificial Intelligence, with its capability to process vast data sets and offer real-time insights, beckons the construction sector into a new age of safety management.

2.2. Artificial intelligence

Artificial intelligence (AI) has significantly impacted the construction safety domain, with various AI technologies including machine learning (ML), Computer Vision (CV), Natural language processing (NLP), knowledge-based systems, and robotics. An understanding of these technologies sets the foundation for discussing specific AI applications in construction. It is important to note that these technologies are not used as a framework for categorizing the applications in this review. Instead, our categorization is based on the type of input data each application uses, as this often dictates the AI methodology employed. This section serves to familiarize the reader with the terminologies and concepts that will be encountered throughout the review, providing a foundational understanding of the AI landscape in construction safety. Additionally, we acknowledge that AI encompasses a broad and evolving range of technologies.

Artificial Intelligence as defined by Rich and Knight “the study of how to make machines do things, which at the moment, people do better” [27]. Artificial Intelligence (AI) is categorized into three primary forms. Artificial Narrow Intelligence (ANI) is specialized in singular tasks like translating languages or forecasting weather. Artificial General Intelligence (AGI) mirrors human intelligence, enabling machines to handle diverse tasks autonomously. The foundation of AI is built upon six components: (1) Learning, the assimilation of knowledge through experience, (2) Knowledge representation, the organization of information in machines, (3) Perception, interpreting sensory inputs, (4) Planning, decision-making based on information, (5) Action, executing tasks from decisions, and (6) Communication, enabling interaction with humans and machines. These facets drive AI advancements in areas like machine learning and robotics.

Machine Learning is an AI subset dedicated to crafting algorithms and statistical models that let computers learn from data without being explicitly programmed. The essence of ML is to identify patterns, predict outcomes, and enhance performance through iterative learning. ML encompasses various techniques: supervised learning (learning from labeled data), unsupervised learning (unlabeled data), and reinforcement learning (reward-based decision-making).

Computer Vision equips machines with capabilities resembling human vision. Through specialized algorithms, it processes visual

information, identifies objects, contextualizes scenarios, and detects anomalies. In construction, computer vision is used to automate various tasks such as detecting hazards [32], inspecting structures for quality issues [33], tracking people and assets on site, and performing earth moving calculations [19].

Natural Language Processing enables machines with the ability to understand and generate human language. NLP enables tasks like sentiment analysis, machine translation, and text generation, making human-computer interactions more intuitive. LLMs in the construction industry have the potential to analyze extensive datasets, identify safety concerns from documented incidents, and predict potential risks. These models can craft safety guidelines, evaluate worker feedback, and generate safety briefings customized to specific construction situations.

Knowledge-Based Systems utilize domain-specific knowledge to assist in decision-making. These systems imitate human expertise through a structured knowledge database. Techniques within this domain include rule-based systems, case-based reasoning, and fuzzy logic methods, among others [27].

Robotics reinforced by AI foundations such as ML and CV, are revolutionizing task automation, emphasizing precision. In construction, robots can do repetitive and hazardous tasks, allowing humans to focus on more complex duties. Robots facilitated by AI, can inspect job sites, transport materials, and contribute to foundational construction, enhancing safety [34]. Moreover, AI-powered robots can proactively detect safety hazards, informing workers of potential risks, especially in collaborative scenarios [32].

3. Review methodology

The review of AI applications in construction was completed in three phases: 1) Identification of literary sources, 2) Filtering and selection of relevant research articles, and 3) Literature coding and Analyses as described below.

3.1. Identification of literary sources

The review process began with the identification of relevant academic journals that publish latest research articles on AI based construction safety automation. Journals were selected based on the various parameters such as impact factor, cite scope, Scientific Journal Rankings (SJR) and popularity in the field of construction. Web of science, and Scopus were used as a primary search engine. Table 1 shows the distribution of reviewed articles from each journal.

Relevant keyword parameters were set to confine the search to the research articles that utilized artificial intelligence to automate construction safety management processes. The keywords were determined by analyzing frequent words found in papers where AI was applied to enhance construction safety. Initially Google Scholar was used to find the papers where text, visual, and audio data are used to automate safety management. Frequent keywords are determined after analyzing the papers retrieved from the Google Scholar. To find articles within the scope of the keywords, Web of Science and Scopus were used. Three sets

Table 1
Journals and articles.

Journal title	Number of articles
Automation in Construction	59
Journal of Computing in Civil Engineering	16
Advanced Engineering Informatics	16
Journal of Construction Engineering and Management	12
Safety Science	06
Computer-Aided Civil and Infrastructure Engineering	04
Engineering, Construction and Architectural Management	03
Applied Sciences	03
Conferences and Other Journals	34
Total	153

Table 2

Keywords by data types.

Data types	Keywords
Textual data	"Text mining" OR "text documents", OR "Natural language Processing", OR "safety risk prediction" OR "classification" OR "narratives" AND "construction", OR "health and safety", OR "risk assessment", OR "hazard", OR "accident", OR "incident", OR "safety" OR "Injury reports"
Visual data	"Computer vision" OR "object recognition", OR "object tracking", OR "action recognition" AND "construction", OR "health and safety", OR "risk assessment", OR "hazard", OR "accident", OR "incident", OR "safety"
Audio data	"Sound detection", OR "sound monitoring" OR "automated sound detection" AND "construction", OR "automation" OR "health and safety", OR "risk assessment", OR "hazard", OR "accident", OR "incident", OR "safety"

of keyword attributes are delineated for each type of input data. These attributes are divided into primary and secondary categories. For instance, primary keywords encompass 'text mining', 'text documents', and 'Natural Language Processing' for textual data; 'computer vision', 'object recognition', 'object tracking', and 'action recognition' for visual data; and 'sound detection', 'sound monitoring', and 'automated sound detection' for audio data. These are used in conjunction with secondary keywords such as 'construction', 'health and safety', 'risk assessment', 'hazard', 'accident', 'incident', 'safety', 'injury reports', and 'automation'. These common keywords across all input data types are specifically chosen to refine the search to studies pertinent to construction safety and health management. The Boolean logic operator 'AND' is employed to search for combinations of primary and secondary keywords within the 'title', 'abstract', or 'keyword' fields of journal articles, aiming to encompass a broad spectrum of sources relevant to the application of AI in construction safety management. Table 2 lists the keywords that were used for literature search. Fig. 1. Illustrates the keyword occurrences of Artificial Intelligence and construction health and safety automation. The networks were mapped by their weights and number of occurrences in research articles along with the association of different research areas of Artificial intelligence.

3.2. Filtering and selection of relevant research articles

The search resulted in more than 1800 search results that were examined, and the relevant articles were selected first using title screening and then using the inclusion and exclusion criteria outlined in Table 3. The inclusion and exclusion criteria were established to ensure a focused selection of research articles within the construction safety domain. This approach aimed to include works specifically utilizing text, visual, and audio as primary data sources to automate construction safety related tasks using AI. The criteria also aimed to narrow the selection to articles published within a specific timeframe (2013 to 2023) and within reputable peer-reviewed journals, enhancing the reliability and relevance of the chosen studies. Some articles that were missed during the keyword search process were identified from using reference chains. Finally, 153 relevant articles were reviewed for this paper.

3.3. Literature coding and analysis

Literature coding began with categorizing the retrieved 153 research articles by three data modalities (i.e. text, visual, and audio). To analyze the article from each data modality, the articles within each modality are further classified by the key tasks. Two key questions are asked to further classify the articles, 1) what safety tasks are accomplished using AI? and, 2) what algorithms/techniques are implemented to achieve that task? These questions helped further categorize the literature based on the safety tasks and key techniques and algorithms used. For instance, texts are generated in construction in the form of project reports, logs, safety compliance documents, and incident reports. Therefore, it is vital

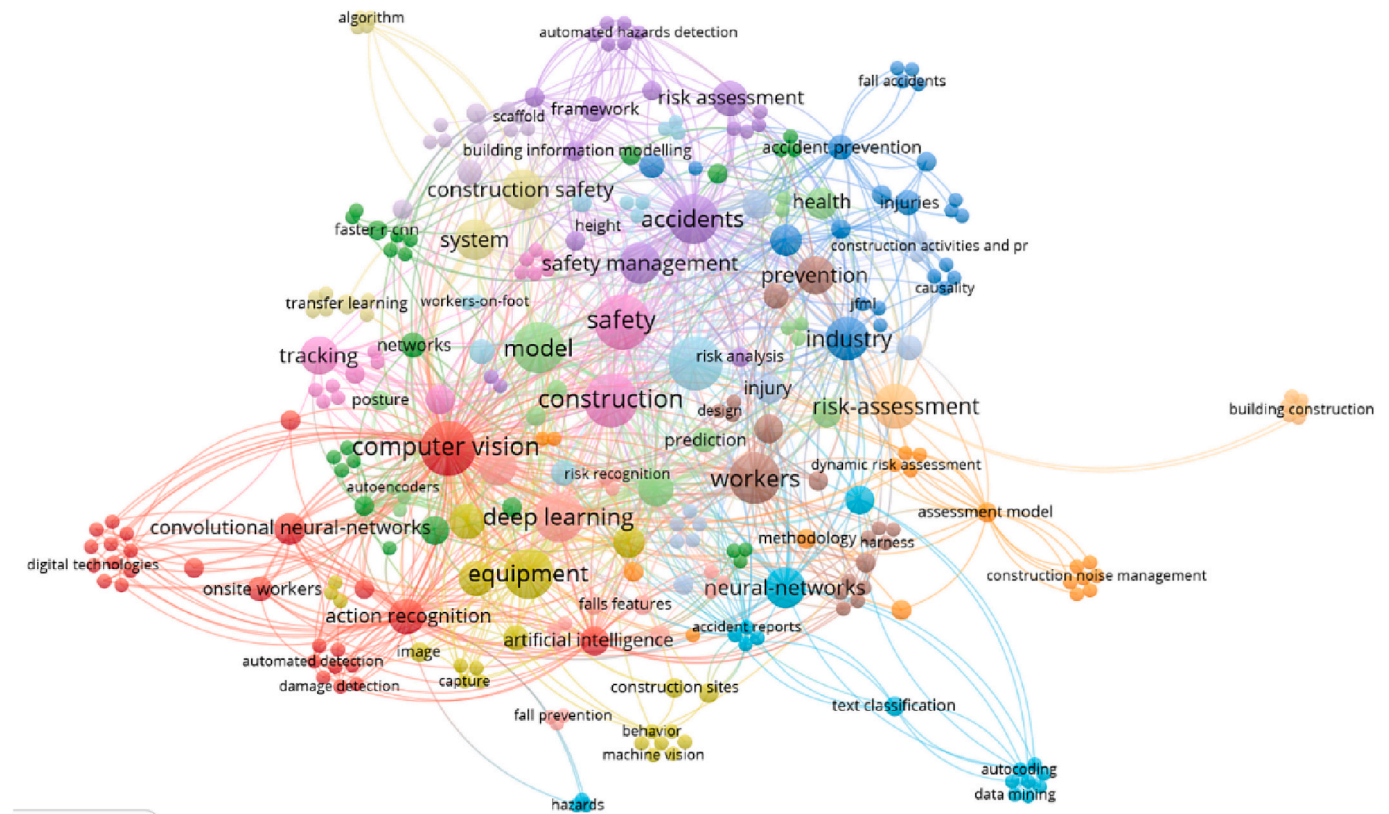


Fig. 1. Keyword co-occurrence network.

Table 3
Inclusion and exclusion criteria.

Inclusion Criteria	Exclusion Criteria
Research articles in the construction safety domain.	Manuscripts focusing on areas other than construction
Use of images/ videos, text and audio as the primary source of data.	Use of AI in design, asset management, real-estate sales and other non-construction activities within the construction field.
Developed/ used one or more AI techniques for a construction safety related task.	Manuscripts published outside the time frame of 2013 to 2023
Work published in journals grouped in the first quartile of SJR rankings.	Manuscripts from non-peer-reviewed conferences and Industry white paper

to retrieve key information, classify similar incidents, automate analysis process and outcome predictions to automate safety management process. On the contrary, visual data has the extended capabilities to extract two-dimensional construction scenes from the visual sensors. Therefore, the safety tasks include object recognition, tracking and various activity recognition in construction.

After categorization, the articles in each sub-category were thoroughly analyzed to obtain 1) Techniques utilized and their effectiveness: the review identified traditional machine learning to advanced deep learning algorithms such as Support Vector Machine (SVM) Deep Convolution Neural Networks (DCNN), Recurrent Neural Networks (RNN), Graph Neural Networks (GNN), and transformer-based attention models. These techniques were applied to tasks such as hazard detection, incident reporting, safety compliance monitoring, and unsafe behavior recognition. 2) Goals and objectives: The primary goal across the studies was to enhance construction safety management through the automation of monitoring and analysis processes. Objectives included reducing accident rates, improving hazard identification, and automating the extraction of actionable insights from safety reports. 3) Benefits and challenges: the benefits highlighted in the studies include increased efficiency in safety monitoring, the ability to process and analyze large volumes of data in real-time, and the potential to significantly reduce human error in safety management practices. Challenges

discussed include the necessity for collecting large, annotated datasets for training machine learning models, semantic feature extraction, and occluded object detection.

4. Review findings

The literature on construction safety automation, spanning from the early 2000s to the present, illustrates a trajectory of evolving methodologies and technologies focusing on text, visual, and audio data analysis. Initially, studies concentrated on text-based analyses, employing traditional statistical methods to investigate accident reports and safety documentation. This foundational work underscored the potential of data-driven insights for enhancing construction safety. As technology advanced, the late 2000s marked the introduction of visual data analysis into the domain. Early computer vision techniques facilitated the detection and classification of safety-related objects and incidents with limitations in accuracy and scalability. The period from 2010 to 2015 witnessed significant advancements in machine learning and computer vision, with the adoption of deep learning algorithms for image recognition and object detection. This era saw the application of convolutional neural networks (CNNs), greatly improving the automation of safety equipment detection and hazardous condition identification. Following the developments, the integration of text and visual data

analytics became more pronounced, leveraging advanced natural language processing (NLP) techniques and transformer-based models for a deeper analysis of textual data alongside visual data, enabling comprehensive insights into construction safety. The most recent phase has expanded into multi-modal data analysis, incorporating audio data processing to detect hazardous noise levels and identify construction activities through sound. This period is characterized by the use of large language models (LLMs) and graph neural networks (GNNs) to enhance data interpretation, aiming for holistic safety monitoring systems that combine text, visual, and audio data. This chronological progression reflects a mature evaluation from basic text analyses to advanced, integrated approaches, demonstrating the field's continuous evolution towards leveraging visual and audio data for advanced construction safety management. The journey from foundational studies to the latest innovations highlights the dynamic nature of research in this area, promising further advancements in safety monitoring and risk management practices in construction. The following sections provide a comprehensive review of the literature for each data modality.

4.1. AI-applications based on text data

Construction industry generates a vast amount of complex data in the form of written documents, such as Minutes of Meetings, accident reports, training records, contracts, progress reports, and correspondence [35]. This textual data is processed using NLP as shown in Fig. 2. Such NLP approaches have the potential to improve construction safety by allowing us to extract useful information from these documents and learn from past accident summaries and injury reports [36].

Among various machine learning techniques, text data mining is widely used to extract structured information from a large collection of unstructured textual data to produce meaningful information and predict future events. In construction safety, NLP can transform unstructured accident data into a structured format, enhancing prediction accuracy through machine learning algorithms. Textual data is first converted into numerical indices using feature extraction techniques such as Bag-of-Words (BoW) and Term Frequency-Inverse Document Frequency (TF-IDF) [37,38]. Unlike language representation models, TF-IDF is a statistical measure used to evaluate the importance of a word in a document, within a corpus, by reducing the weight of terms that occur frequently across documents, thereby highlighting more distinctive terms in each document. The integration of deep learning into NLP

further augments the utilization of text data for construction safety management. Recent research has shown that most AI applications involving text data revolve around three major categories, which are automatic information extraction from accident reports, retrieval of similar accident reports, and safety outcome prediction. Table 4 outlines studies employing text-based AI for construction safety, detailing commonly used algorithms. Subsequent sections delve deeper into these applications.

4.1.1. Automatic information extraction from accident reports

Accident reports include valuable information on the details and causes of past accidents and analyzing this information can help improve safety and health performance. A considerable body of scholarly investigations leverages text mining and Natural Language Processing (NLP), employing various feature extraction techniques (e.g., BoW, TF-IDF), rather than complex language models, to process and categorize accident narratives. This approach enables the automation of information extraction and classification from accident reports, offering insights into the root causes of incidents. With such processed data, construction safety professionals can develop an empirically informed safety approach to mitigate future risks. For example, several studies have demonstrated the evidence of implementing a hybrid approach, stacking multiple algorithms to classify relevant information from accident narratives [35,39]. Besides, [39,40] have also evaluated various machine learning classifiers to extract and categorize meaningful information ranging from support vector machine (SVM), linear regression (LR), random forest (RF), k-nearest neighbor (KNN), decision tree (DT), to Naive Bayes (NB) using inputs processed through TF-IDF. This demonstrates the utility of TF-IDF as a feature extraction method that complements various machine learning algorithms in analyzing textual data. The studies found that Support Vector Machines (SVM) have the highest accuracy for classifying accident narratives. In addition, one of the studies evaluated the strength of unsupervised learning in classifying the causes of accidents in Arizona using OSHA accident reports [41]. However, some major shortcomings were found in terms of weak learning performance and error rate. To address the shortcomings, [42] proposed a hybrid supervised learning model, Symbiotic Gated Recurrent Unit (SGRU), integrating state-of-the-art recurrent neural networks that includes Gated Recurrent Unit (GRU) and Symbiotic Organisms Search (SOS). The results indicate that SGRU is the best classification algorithm compared to other AI models. However, recent experiments

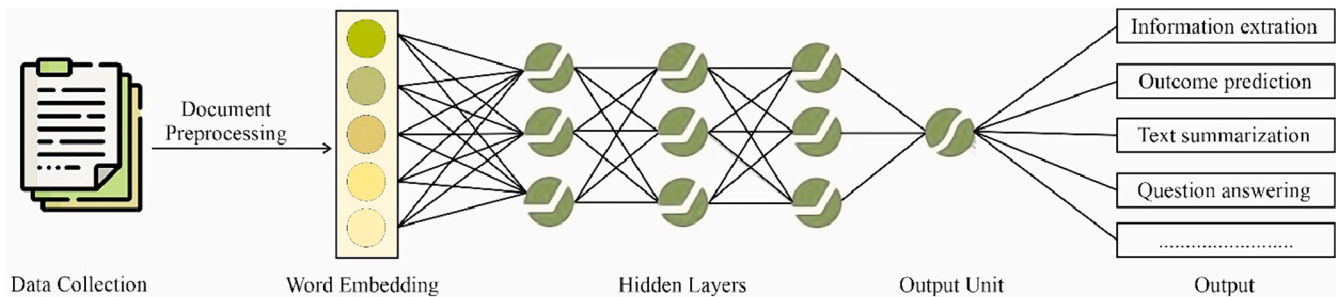


Fig. 2. Generic pipeline of NLP.

Table 4
Journals of text-document based construction safety.

Task	Algorithm(s)/classifier(s)	References
Automatic information extraction from accident reports	BoW/ TF-IDF / k-NN, RF, LR, SVM (Hybrid model)/ RNN/Symbiotic Gated Recurrent Unit (SGRU)/ Unsupervised clustering	[35–51]
Retrieval of similar accident reports	BoW+TF-IDF/ Case Based Reasoning (CBR) / Conditional Random fields (CRF) / Okapi BM25/ Word2Vecs	[52–54], [56–62]
Automatic accident report analysis and safety outcomes prediction	BoW/ Decision Tree/ RNN/ CNN /HAN/C5.0 and CHAID	[63–65], [67–79]

have emphasized more on embedding LSTM networks after a deep neural network while classifying safety reports based on the hazard attributes, as the additional recurrent network provides more contextual information [43–45].

Recent contributions to the construction safety domain have introduced novel methodologies. For instance, Xu and Zou (2021) delved into a vast dataset of workplace injuries spanning 11 years, unraveling key insights on injury patterns and causes [46]. Similarly, Trillo Cabello et al. (2021) worked with a dataset of accidents in Spain, extracting knowledge from historical data using association rules. Their findings offer invaluable insights poised to reshape safety protocols in construction [47]. Furthermore, Rupasinghe et al. (2021) harnessed the capabilities of Text Mining and NLP to extract hazard sources from public injury reports. Their rule-based tool showcased superiority over conventional statistical classifiers, emphasizing the worker's pivotal role in accident occurrences [48]. Progressing chronologically, Feng and Chen (2021) introduced a data augmentation technique tailored for small sample sizes, highlighting that the effectiveness of AI doesn't solely hinge on vast datasets [49]. Venturing into more recent advancements, Wang et al. (2023) adopted Named Entity Recognition to derive fall protection requirements from safety regulations, showing commendable precision and recall metrics [50]. Additionally, Wu et al. (2023) developed the knowledge representation models using the BERT-BiLSTM-CRF model and demonstrated remarkable precision and recall rates underscoring the potential of this robust approach [51].

4.1.2. Similar accident case retrieval

Similar accident case retrieval from large accident databases of handwritten and text-based unstructured documents has been a huge obstacle that prevents learning from previous accidents. Numerous studies have recommended safety professionals to learn from past accidents for better risk assessment during the initial phase of a construction project [52,53]. However, the traditional retrieval process implements binary search algorithm that only searches for the documents that are similar or different by the given queries. Previous accident cases that might be worth conducting the analysis, are ignored in this method due to the wide range of synonyms used in construction domain. The potential of artificial intelligence can be used in this case to extract semantic accident documents [54] by using various machine learning techniques and case-based reasoning methods.

Cased-Based Reasoning (CBR) is a problem-solving approach that leverages previous experiences stored in a database of past events [55]. To encourage the widespread use of CBR, efforts have been made to create databases of accident cases, such as the National Institute for Occupational Safety and Health (NIOSH), which has developed a database consisting of a range of cases to support safety management [56]. Similarly, Zhang et al. created a risk case database of 249 incidents to improve risk management in the Shanghai metro area [57]. However, traditional risk case databases are often composed of informal and unstructured written documents, making manual evaluations time-consuming and labor-intensive. To address this, researchers have applied CBR and Natural Language Processing (NLP) techniques to efficiently retrieve similar accident reports from the database. For example, Zou et al. developed a framework that uses two NLP techniques (Vector Space Model and semantic query expansion) to retrieve similar cases [52]. Similarly, Kim and Chi proposed a similar model that also uses information extraction methods to analyze the implicit knowledge contained in construction accident reports, with Conditional Random Field (CRF) used for automatic knowledge extraction [58]. The accident cases are ranked, and weights are calculated according to the thesaurus. Another approach used web crawling and the Word2Vec vector space model to collect data from websites and determine similar terms from the reports [59].

Recent developments include the introduction of hybrid deep learning models. Specifically, a model developed by Zhang et al. (2022) effectively employs Word2Vec for construction accident cause

classification, demonstrating superior performance over traditional models [43]. Another notable study in 2023 by Tian et al. introduced an intelligent question-answering method for construction safety, employing deep semantic mining [60]. Additionally, Alkaissy et al. (2023) demonstrated the capabilities of machine learning in classifying injury types using extensive construction accident records from Australia [61]. Building on this, both Luo et al. and Zhong et al. in 2023 introduced deep learning models for accident report classification and hazard analysis, respectively. These contributions further underscore the growing influence of advanced AI techniques in the field [53,62].

4.1.3. Automatic accident report analysis and safety outcome prediction

Automated accident report analysis and safety outcome prediction are crucial for safety management and can significantly streamline the safety management process. As a result, attribute-based construction safety outcome prediction has been suggested by the researchers, applying both NLP and machine learning techniques [63,64]. Recently, Bakera et al. proposed a robust deep learning model that automatically learns injury precursors from raw accident reports using Convolutional Neural Networks (CNN) and Hierarchical Attention Networks (HAN) algorithms [65]. Similarly, Zhonga et al. proposed a hybrid deep learning model that uses Latent Dirichlet Allocation (LDA) [66] to automatically identify hazard topics from accident reports and CNN algorithms to classify the hazards [53]. Both models utilize state-of-the-art deep learning tools to automate the hazard identification and prediction process in the field of construction research. Additionally, some studies have applied simple data mining techniques, such as decision tree algorithms (e.g., Random Forest, C5.0, and CHAID), to model accident data obtained from the accident sites [67,68].

Supplementing these techniques, Wang and El-Gohary introduced a deep learning methodology for extracting and representing safety requirements related to fall protection in construction, harnessing the power of convolutional neural networks and knowledge graphs [69]. Also targeting efficient analysis, Pan et al. developed a graph-based deep learning framework, using the Graph Convolutional Network, to classify accident reports, spotlighting accident causes and their impacts [70]. Diverse machine learning models have expanded the horizons of prediction and management in the construction safety domain. For instance, Zermane et al. employed the Random Forest model to unearth factors leading to fatal falls in Malaysia's construction sector [71]. Delving into the challenges of imbalanced datasets, Koc et al. utilized machine learning methods paired with resampling techniques to predict accident outcomes, underscoring the influence of attributes such as past company accidents and a worker's age [72]. The same year, the team introduced an ensemble model identifying susceptible body parts of construction workers before accidents, emphasizing influential attributes like worker age and company size. By 2023, the authors progressed to creating a national data-driven framework for construction safety, using a vast dataset to predict fatal accidents, incorporating the SHAP approach for enhanced model interpretability, and devising a safety management plan based on these findings [73,74]. Meanwhile, Salarian et al. merged clustering and oversampling to tackle heterogeneous and imbalanced datasets, enhancing the precision of ensemble models, including Random Forest [75]. Encompassing broader frameworks, Gondia, et al. introduced a comprehensive machine learning framework designed to bolster decision-making processes, focusing on injury risk mitigation [76]. In an innovative approach, machine learning based Bayesian framework was integrated to predict construction incidents using interval estimates for proactive decision-making and risk management in the construction industry [77]. Notably, in 2023, graph convolutional networks were employed to enhance risk assessment by linking related accidents [78]. Moreover, an automated system was introduced to streamline the selection of optimal machine learning methods for construction site safety [79]. These advancements emphasize the potential of text data-driven AI applications to revolutionize construction safety management.

The application of AI in construction safety management, particularly in automatic information extraction from accident reports and similar accident case retrieval, has brought about profound implications. By automating the analysis of accident reports, AI provides invaluable insights into the causes and details of past accidents, allowing safety professionals to develop more effective risk mitigation strategies. This approach is pivotal in learning from past accidents to enhance future safety measures. Similarly, the use of AI in retrieving similar accident cases from large databases addresses the challenge of learning from previous incidents, which has been hindered by traditional retrieval methods. Recent advancements in AI enable a more comprehensive and nuanced understanding of accident causes and patterns. The shift towards AI-driven analysis and prediction in construction safety management is revolutionizing the field, offering a more proactive and data-informed approach to ensuring safety in construction projects. It highlights the shift from manual, labor-intensive methods to efficient, AI-powered strategies, fundamentally changing how safety is approached and managed in the construction industry.

4.2. AI-applications based on visual data

Computer vision has become an important tool for the construction industry in recent years. By leveraging advanced ML and deep learning

algorithms, it has the potential to transform construction safety management. For instance, computer vision can monitor construction sites in real-time, identifying hazards such as workers without protective gear [80], heavy or dangerously positioned equipment or workers in undesignated areas [81]. CV fundamentally entails preprocessing raw images, training them with preferred machine learning techniques, and deploying the tested model for real-world inferences (Fig. 3).

Various machine learning algorithms can be used to analyze images and extract information related to construction safety. These include traditional machine learning approaches, such as decision trees and support vector machines. Deep learning methods, especially CNNs, have surpassed traditional methods by automating feature extraction and excel in tasks like object recognition. Table 5. lists common computer vision tasks that are used for safety applications along with some popular algorithms used in these applications. The following sub-sections discuss these applications in detail.

4.2.1. Object recognition

Object recognition in computer vision refers to the process of recognizing the presence, location, and shape of objects in an image or video, and then classifying them based on their features such as color, texture, and shape. Object recognition has many practical applications, including construction safety, where it is used to detect potential safety

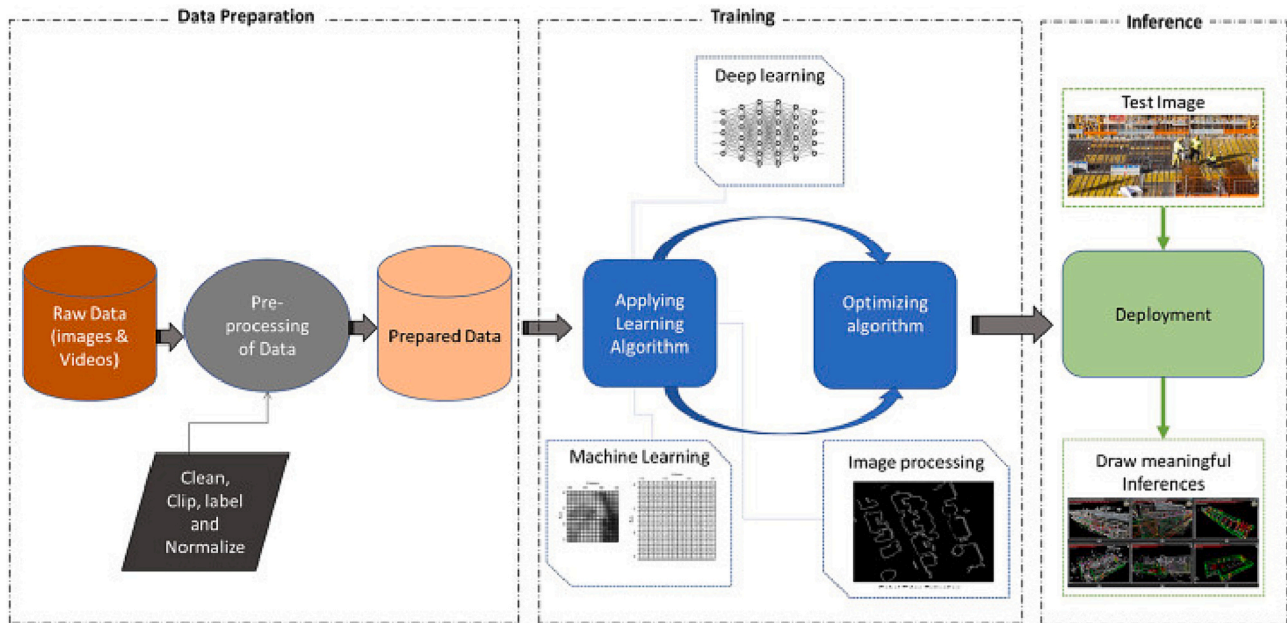


Fig. 3. Typical computer vision process (Adapted from [19]).

Table 5

Various computer vision tasks used in safety management.

Task	Algorithm(s)/classifier(s)	Literature
Object recognition (Mobile heavy equipment; workers; dump trucks; excavators; hardhat; loader, concrete mixer truck, and road roller; safety harness & guardrail)	SVM/ K-NN / HAAR, HOG, HSV color histogram R-CNN / Faster-RCNN/ Mask-RCNN/ YOLO/ CenterNet/ C-BERT	[80–82], [88,93,100–102] [32,83,84,87,89,90,92,94,95], [103–125]
Object tracking (Multiple workers; wheel loaders and trucks; excavators; roller and dozer)	Silhouette tracking algorithms/Probabilistic Hausdorff image matching Kernel tracking/ Kernel co-variance tracking algorithms Point tracking/ Eigen-image/ Particle filtering/ median flow and pyramidal Lucas-Kanade algorithm Hybrid 2D tracking algorithms 3D tracking algorithms	[131] [126,128,132] [127,129,133] [130], [134–139]
Action recognition (Actions of workers, excavator, ladder climbing, trucks, Liner-lowering, tower crane and dumping waste from heights)	Faster R-CNN, RNN, Hidden Markov models (HMM), Conditional random fields (CRF), BoW, KNN, SVM	[141–160]

hazards and monitor the construction site for safe conditions [82]. Popular models for object recognition include Region-based Convolutional Neural Networks (R-CNNs) and their variants are widely popular for safety applications due to their high accuracy [83,84]. However, there are also other modern object detection models that prioritize speed over accuracy, such as You Only Look Once (YOLO), Single Shot Detector (SSD) [85], RetinaNet [86]. These algorithms offer fast and efficient object detection, proving to be useful for real-time safety applications.

Construction safety researchers have demonstrated the potential of object recognition to identify safety non-compliance issues with high accuracy ranging from detecting PPEs [87], Hard-Hat [88], standard guardrails [89], to structural supports [90]. Alternatively, object classification technique has also been used as a part of object recognition method to classify construction workers [91], risk zones [92], equipment [93], safety behaviors [94], and safety hazards [95]. In addition, research in vision-based safety management has also focused on developing and evaluating different algorithms from conventional detection algorithms such as Feature Transform (SIFT) [96], Histogram of Oriented Gradients (HOG) [97], Haar-like features [98], and the Speeded Up Robust Features (SURF) [99] to most advanced deep learning-based algorithms. Azar, McCabe [100] and Park, Brilakis [101] experimented with feature-based algorithms to detect construction dump trucks and equipment respectively. Likewise, Zou and Kim developed a simplified approach utilizing color-based features integrating HSV color space model to extract excavator of interest from the construction visual data captured from a foundation excavation activity [102]. The study reports that due to the lack of adequate sunlight, HSV might fail to detect construction objects at night.

In contrast to traditional feature-based methods, deep learning algorithms autonomously extract essential features from visual data, as demonstrated by Kim et al. using the R-FCN classifier for on-site heavy equipment identification [103]. The Faster R-CNN emerged as a leading object recognition technique in construction safety, detecting non-hardhat users [83], workers [104], nails and screws [105]. In addition, previous studies also explored workspace recognition and worker trade identification [106,107]. Recent innovations merged mixed reality with object detection for hazard identification [108] and eye-tracking was also combined with vision-based hazard recognition frameworks [109], including SLAM algorithm for workers' localization [32]. Progressing into the year 2020, technologies utilized object detection combined with ontology for advanced hazard detection [110,111].

By 2021, there were significant strides in object recognition. Shen et al.'s method achieved a 94.47% accuracy in safety helmet detection using deep transfer learning [112]. Combining computer vision and ontology improved construction safety inspections [113], meanwhile innovative techniques emerged for construction vehicles detection using point cloud [114] and missing barricade detection using U-Net and CenterNet [115]. As we transitioned into 2022, Goh et al. introduced a real-time system employing CenterNet for worker detection, ensuring safe distancing protocols during the COVID-19 era [116]. Deep learning models such as YOLOv5 enhanced helmet detection under various conditions [117,118], while other systems focused on fall prevention [119], and continuous safety compliance monitoring [120]. Also notable, Zhang et al. combined construction scene graphs with a C-BERT network, achieving an astonishing 97.82% hazard identification accuracy [121], and Chian et al. devised a dynamic method for crane load fall zone detection [92].

The trend of integrating traditional methods with deep learning techniques continued into 2023. Seul-Ki Lee and Jung-Ho Yu combined conventional AI image recognition with relation-based reasoning, presenting an ontology for construction site hazards [122]. This fusion led to the discernment of hazards on construction sites even without extensive training datasets. Park et al. introduced SOC-YOLO for enhanced worker detection under challenging conditions [123]. Addressing construction accidents, Xu and Wang presented a predictive

model with an active warning mechanism [124], while Chen et al. showcased a model enhancing safety monitoring, especially under occlusion scenarios [125].

4.2.2. Object tracking

Object tracking is another major research topic in the field of computer vision with significant potential for enhancing construction site safety. By tracking construction workers [126], equipment [127], site dynamics [128], and onsite objects [129], object tracking systems can help ensure the safety of workers by monitoring their movements and relative distances using computer vision-based object detection. The objective of object tracking is to consistently locate a moving object of interest with an embedded unique ID across a series of video frames. The object of interest can be represented by shapes (e.g., skeleton, points, geometric shapes, and contours) and appearances (e.g., color, edges, and texture), and a temporal trajectory of the moving objects can also be created as they move around the scene.

In the context of construction safety, object tracking primarily focuses on construction workers and project entities (e.g., construction equipment, vehicles, objects, and resources) to ensure a safe distance from hazards [130]. Such tracking in computer vision primarily relies on 2D-based object tracking of construction elements [128]. Three predominant methods for 2D object tracking include Silhouette tracking [131], kernel tracking [132], and point tracking [133]. Silhouette tracking, popular in construction, uses color histograms, object edges, contours, or silhouettes to represent objects, making it adept at depicting complex shapes [129]. On the other hand, kernel tracking computes object motion across video frames, proving reliable even under variable illumination. To the end, Point tracking represents objects as points within video frames, effective in handling occlusions.

However, Konstantinou et al. (2019) proposed a more refined 2D tracking method with adaptive, predictive, and appearance models tailored for dynamic construction setting [134]. Due to the limitations of 2D methods, researchers turned to 3D tracking. Such techniques yield 3D position, direction, and speed of objects, demonstrating proficiency in monitoring construction equipment. For instance, Lee and Park (2019) utilized stereo vision to track 3D positions of workers [135], while Cai and Cai (2020) blended vision-based tracking with radio-based identification to enhance accuracy [136]. Similarly, Neuhausen et al. (2020) employed synthetic data to boost tracking performance [137]. In 2021, Xiao and Kang introduced a deep learning method, the Construction Machine Tracker (CMT), showcasing remarkable precision in multiple construction settings [138]. Adopting a forward-thinking approach, Kong and Fang (2021) combined computer vision with LSTM to forecast unsafe construction behaviors, marking a progressive use of tracking for safety [139]. While advancements in vision-based tracking are noteworthy, challenges persist, such as occlusions, scale variations, and abrupt movements. Nonetheless, computer vision surpasses sensor-based approaches like RFID, UWB, and GPS, as it does not necessitate sensors on project entities and provides expansive site coverage.

4.2.3. Action recognition in construction safety management

Action recognition is the most dynamic computer vision task that involves both object recognition and tracking to provide a comprehensive understanding of an environment. The purpose of computer vision-based action recognition system is to determine the action of one or more agents of interest in a series of images or video streams. The process of action recognition process is mainly initiated by extracting features and correctly classifying them into the predefined action categories from the consecutive images or video frames. Some common algorithms that were previously implemented to automate action recognition tasks in safety management are Faster R-CNN, RNN, Hidden Markov models HMM, CRF, BoW, KNN, and SVM.

In the construction industry, a notable number of accidents stem from complex interactions between workers and equipment. While

object recognition and tracking provide contextual and positional information, action recognition discerns the movements of workers and machinery. Recognizing the actions and movements of both workers and machines is also critical. A detailed examination of human behavior which includes assessing postures, gestures, actions, interactions, and overall activities, is crucial for detecting unsafe behavior on construction sites. For example, the Occupational Safety and Health Administration (OSHA) highlights the necessity of ergonomic postures during the handling of materials to minimize the risk of injury [140].

Recent advancements in computer vision emphasize the role of action recognition in construction safety. Beginning with the innovation by Gong et al. (2011), an automated classification system for construction workers and equipment actions was introduced, utilizing a Bag-of-Video-Feature-Words and a Bayesian network to adeptly handle occlusions and viewpoint changes [141]. Researchers furthered this trajectory with a system to track and quantify dirt-loading cycles in earthmoving operations [142]. The subsequent years observed significant contributions that leveraged motion feature analysis for the classification of unsafe actions with high accuracy [143], as well as vision-based motion capture and recognition frameworks to monitor worker behavior [144]. A study in the following year introduced methods for activity analysis using depth images to assess work in interior construction operations with notable precision [145]. The same year, yang et al. (2014) proposed a framework for tracking tower crane activities that effectively predicted activity states from video footage [146]. The development further outperformed existing methods using dense trajectories to improve recognition accuracy of the workers from construction videos [147]. By 2017, advancements had been made in real-time identification of unsafe behaviors using image-skeleton-based methods [148] and semantic annotation of videos for better data utilization on construction sites [149]. The momentum continued following years with the integration of deep hybrid learning model and ergonomic posture recognition framework using 3D view-invariant features which marked a step change in detecting and assessing workplace safety [150,151]. The former studies were complemented by the application of the Faster R-CNN to classify construction activities, which opened avenues for more nuanced and rule-based activity categorization [152]. The emergence of neural networks tailored for excavator manipulator pose estimation [153], alongside marker-less pose estimation systems for construction robots, and the utilization of sequential pattern analysis for action recognition [154], collectively pushed the boundaries of automation and productivity monitoring in the construction sector.

Overlaying these individual developments, from 2020 to 2023, the integration of deep learning techniques has become increasingly prominent for real-time safety monitoring, such as the applications of Faster R-CNN and YOLOv5-FastPose models for assessing machinery and worker poses [155,156]. In recent advancements, risk identification analyzing the vibrations in scaffolding structures [157], 3D skeleton and LSTM model for emergency notification interpreting hand gestures [158], the Spatial Temporal Relation Transformer to detect safety behaviors [159], dual-channel C3D models and detection of dynamic 3D hazardous proximity zone around excavators [160], have substantially contributed to the field's advancement.

4.3. AI-applications based on audio data

Although audio data-based AI technologies are underexplored in construction safety research compared with text and vision-based methods, opportunities exist to use audio data to detect hazard in complex and vulnerable construction environments. The process of audio classification and recognition is comprised of extracting various audio features from sample audio data and feeding them into a machine learning algorithm to identify predetermined sound classes (Fig. 4). Such AI systems can effectively analyze and derive critical information from the auditory outputs of construction activities, including excavation, drilling, rolling, lifting, and piling.

Auditory based construction safety surveillance system combined with machine learning techniques have numerous economical solutions to offer for the critical issues encountered in construction safety management. For instance, one of the severe issues in construction safety management is lack of sound sensing of the construction workers, that can result in one of the common accidents relating to “struck by moving vehicles”. In response to solve the proximity related hazards, Dang et al. (2020) previously developed a novel audio-based machine learning model for automated detection of collision hazards in construction sites [161]. The study incorporated a CNN based deep learning model for automated sound detection of collision hazards with a new labeled dataset of normal and abnormal sound events. Consequently, a recent study has proposed a mobile device based struck-by accident hazard recognition module by analyzing Doppler effect caused by the movements construction workers and equipment using Convolutional Neural Network (CNN) [162]. Building on this foundation, Elelu et al. (2023) have recently proposed an affordable technology that uses audio signals to detect mobile equipment in construction sites, with the primary goal of enhancing workers’ auditory awareness in noisy environments [163]. Adding another layer to this auditory surveillance system, Kim et al. have developed construction sound localization framework [164]. A pivotal finding from their research is the system’s ability to discern between accidental and machinery-related sounds, a capability that holds paramount importance for both safety and productivity. Similarly, CSafe, a wearable audio platform presented by Xia et al. (2021), focuses on enhancing construction worker safety by detecting and localizing approaching vehicles in noisy construction environments [165]. With a unique sound source separation algorithm, CSafe demonstrated a detection rate of 82%, outclassing other comparable safety technologies. Similarly, a study has also demonstrated a machine learning based construction sound identification framework that can identify safety hazards and at the same time notify construction workers in advance [166]. Likewise, a previous investigation also proposed a similar audio-based autonomous safety surveillance system with prenotification alarm, and integrated map visualization feature with it [167].

Meanwhile, Scarpiniti et al. (2021) proposed two distinctive approaches for audio classification in construction sites. One introduced a Deep Belief Network (DBN) achieving a classification accuracy of up to 98%, and another deployed a Deep Recurrent Neural Network (DRNN) with LSTM units, which showed an accuracy rate of up to 97% [168]. In the subsequent year, they also tackled the challenge of generating audio data in construction sites with the introduction of the CoVal-SGAN

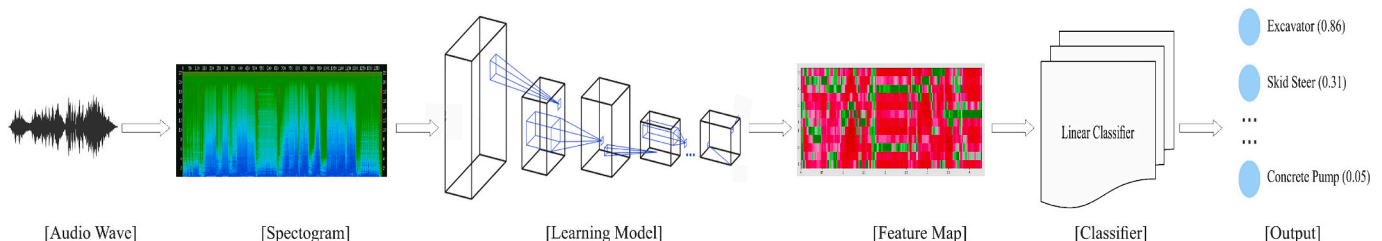


Fig. 4. Overall process of audio classification.

model, which showcases improved accuracy against other methods. Recent research by Xiong et al. (2022) further deepens the domain, introducing a method that efficiently handles overlapping construction noises and identifies specific activity timings using a blend of CNNs and RNNs [169]. Moreover, Akbal et al. (2022) have brought forward Des-PatNet25, a unique system for monitoring construction sites with sound signals that employs the data encryption standard's first S-Box as a feature generator [170].

Audio-based surveillance systems, earlier developed for monitoring construction work activities and equipment operations to enhance productivity [171–173], also have efficacy towards ensuring safety. For example, hybrid systems were also found efficient in recognizing multiple construction equipment and activities proposed by [174,175]. However, the acquisition of comprehensive datasets remains a challenge due to the varied nature of construction equipment.

5. Challenges and recommendations for future research

While recent AI applications offer impressive capabilities, there are several challenges that hamper the integration of such systems for safety management applications. These include the necessities for exhaustive, quality-assured datasets and the substantial undertaking of developing algorithms that can adeptly navigate the distinctive volatility and intricacies of construction environments. The following subsections detail different challenges and recommendations for future research opportunities for AI integration for safety management based on the insights derived from the investigation.

5.1. Text-based safety automation

Review identified several challenges in implementing text-based safety automation along with future research recommendations as highlighted in Table 6. These are further discussed in the subsequent sections in detail.

5.1.1. Large scale data collection and annotation

In text-based automated safety research, a pivotal challenge identified across numerous studies is the collection and annotation of data. Text-based automated safety studies predominantly rely upon large scale annotated data. A large-scale annotated datasets of workplace injuries is instrumental in developing machine learning models for accident classification, hazard identification, and safety outcome prediction. While studies mostly suffer from limited access to real-life injury reports, data annotation requires extensive amounts of labor hours from safety experts to manually annotate the text data. In addition, the sensitivity and accessibility of workplace injury reports along with the reluctance of privately owned construction companies to share information, further complicate the issue. Small datasets resulting from these constraints are prone to overfitting in machine learning models, leading to inaccurate predictions. To mitigate these challenges, employing data augmentation

techniques, including the use of transfer learning with pre-trained language models, presents a viable strategy. However, the primary challenge of this strategy lies in its potential to introduce biases or inaccuracies during the data augmentation process, particularly when extrapolating beyond the scope of the original datasets. The advent of Large Language Models (LLMs) offers promising prospects for expediting the annotation process, as evidenced in a study of medical information extraction [176]. The study formulated a method by leveraging both LLMs and human expertise to generate ground truth labels for unstructured medical text annotation. Additionally, exploring innovative approaches such as pseudo-labeling, meta-learning through few-shot and zero-shot learning, and active learning could significantly contribute to overcoming these obstacles in future research endeavors.

5.1.2. Semantic feature extraction

The process of extracting and analyzing semantic features from construction accident reports constitutes a significant challenge with both academic and practical implications. These reports contain a wealth of complex semantic information, such as word sequences, co-occurrences, and nuanced relationships, which are crucial for effective classification and information retrieval tasks. A recent approach employing Graph Neural Networks (GNN) alongside BERT for node embeddings within knowledge graph-based systems has emerged as a promising solution [177]. This approach effectively captures and utilizes semantic relationships, thereby enhancing the analysis and retrieval of relevant information and efficiently handling semantic similarities. The integration of such techniques to distinguish and interpret complex semantic information has the potential to substantially augment the accuracy and reliability of safety management practices, through improved classification and information retrieval. The field of information retrieval has also witnessed significant advancements with the introduction of the Retrieval-Augmented Generation (RAG) technique, which exploits the capabilities of Large Language Models (LLM) [178], further pushing the boundaries of text-based automated safety research.

5.1.3. Handling heterogeneous text data

Handling heterogeneous text data from construction injury and hazard reports in various text formats (e.g., Word, Excel, PPT, CAD drawings, etc.) is another vital challenge observed in previous studies. These types of reports contain critical safety information, yet their diverse formats complicate the extraction and summarization of this data for actionable insights. A study introduces an innovative solution with open source LLM which has been successfully applied in extracting information from different formats of medical texts [179]. This technique, leveraging zero-shot learning and prompt engineering, demonstrates a high degree of accuracy in identifying essential features from unstructured texts of different formats. However, this approach has only dealt with text data while construction safety professionals sometimes might need to extract safety information from graphical data (e.g., CAD drawings). Therefore, LLMs need to have better knowledge representation of construction drawings. Fine tuning a LLM can be a better approach to solve such problems. However, fine-tuning a LLM with such heterogeneous data requires computation resources and time. Recently developed approaches such as Parameter Efficient Fine Tuning (PEFT) [180] with quantization techniques, QLoRA [181] can be viable option to fine-tune LLMs with low computation power and faster inference. Applying these approaches to construction safety documents could revolutionize how data is processed, allowing for the seamless integration and analysis of information across different document types. By adapting the methodologies, safety managers could efficiently extract and summarize critical safety information from heterogeneous sources, thereby enhancing decision-making processes and improving safety outcomes.

Table 6
Summary of the key challenges and recommendations.

Challenges	Recommendations
Large scale data collection and annotation: Manual labor hours and human experts to develop annotated real-life safety datasets.	- LLM to accelerate annotation process
Semantic feature extraction: Poor semantic relation extraction of safety hazards from injury reports	- Graph Neural Networks to improve semantic relationship
Handling heterogeneous text data: Information extraction and summarizing documents from various sources and formats.	- Fine-tuned pre-trained LLMs with PEFT and quantization
Model selection for specific safety tasks: Appropriate NLP techniques for safety related tasks	- HT-SR for model selection

5.1.4. Model selection for specific safety tasks

Selecting diverse ML algorithms for various safety related tasks is also a challenging aspect revealed in the previous studies. Previous studies also experimented with model stacking, ensemble learning, and meta-learning [182]. However, a recent study introduces an NLP model selection approach based on heavy-tail self-regularization (HT-SR) theory [183] which uniquely predicts test error without using training or test data. This contrasts with traditional model stacking and ensemble learning, which are data and computation-intensive, and meta-learning, known for its complexity and task-specific adjustments. The HT-SR method simplifies model evaluation, offering a data-independent metric for assessing pre-trained models, significantly benefiting safety related tasks with limited data access of accident reports while selecting from numerous pre-trained NLP models.

Rapid developments of pre-trained LLMs (e.g., BERT, GPT, LLAMA, and MISRAL) have a huge potential to replace traditional NLP techniques for safety specific tasks in future. Future research endeavors could more focus on the development of safety task-specific LLMs tailored for information extraction, text summarization, question answering systems, and safety bots. Such models could be specifically trained to efficiently extract and retrieve information from injury reports of different formats, enhancing the accessibility and usability of crucial safety data. Another advantage of utilizing pre-trained language models is their ability to address the limitations posed by small datasets. Therefore, it is advisable to employ a range of pre-trained language models at different levels to address the challenges encountered in text-based studies related to construction safety. Additionally, leveraging the capabilities of reinforcement learning for informed safety decision-making presents a promising avenue for advancing the field. By harnessing the potential of these technologies, the construction industry can significantly improve its approach to safety management, potentially reducing the incidence of accidents and enhancing overall workplace safety.

5.2. Vision-based safety automation

Table 7 outlines various challenges in implementing vision-based safety automation and suggests future research recommendations. These topics are elaborated upon in the following sections.

5.2.1. Lack of annotated visual dataset

The limitations of annotated visual data specific to construction environment is one of the most common challenges to train neural networks for unsafe behavior recognition in construction. Weak generalization due to the reliance on small, supervised databases impedes the ability to provide accurate generalizations. The underlying machine learning models, typically using small databases for training, limit inter and intra-class variability. Transfer learning techniques utilizing pre-trained models can partially mitigate the issue. However, the efficacy

Table 7
Summary of the key challenges and opportunities of vision-based safety applications.

Challenges	Recommendations
Lack of annotated visual dataset: The predominant challenge encountered in training neural networks for safety-related tasks	- Generative data augmentation with diffusion models. - Self-supervised image annotation
Occluded object recognition: Accuracy of object recognition suffers from visual obstructions.	- Integration of CompositionalNet with Deep Convolutional Neural Network - GAN based de-occlusion method
Limited object visibility: Real-time safety hazard detection is hampered by limited object visibility.	- Autoencoder based dehazing and denoising
Small object detection: Small hazardous objects are not detected effectively.	- GAN for small object detection

of these methods often shows a propensity towards biases inherent in the specific datasets on which they were originally trained. In addition, The performance of computer vision systems in construction is highly dependent on input data quality. Factors like camera positioning and environmental conditions, such as lighting and obstructions, significantly affect data integrity. Financial barriers related to hardware and staffing for data collection further challenge the adoption of these technologies. Data augmentation stands as a key technique for addressing the issue, enhancing the diversity and volume of training samples. Although conventional data augmentation methods, due to their subjective nature, might introduce distortions or artifacts not reflective of real-world construction environments. This discrepancy can result in models that perform well on augmented data but falter during actual deployment. Recent advancements in diffusion-based models such as Stable Diffusion, Midjourney, Dall-E, and Imagen have enabled the generation of near-infinite variety of photo-realistic images from text prompts. Harnessing the capabilities of diffusion models, a study has demonstrated improved precision in object detection using controllable diffusion models and CLIP [184]. The proposed method involves generating appropriate visual priors to control the generation of synthetic data and implementing post-filtering techniques using category-calibrated CLIP scores. Additionally, this approach has automated the annotation process, significantly reducing the need for manual annotation efforts. Such method offers the potential to create a construction safety-specific dataset, encompassing a wide variety of equipment, viewpoints, and activities. In addition to the limitation of construction specific large visual dataset, annotation and segmentation process of the images is another labor-intensive task which requires expert human annotator that can be automated using Self-Supervised Learning (SSL). SSL is an emerging field that increases performance when unannotated data is present. Recent research highlights substantial performance gains achieved through SSL, suggesting its potential to efficiently annotate and segment datasets tailored to the construction sector [185].

5.2.2. Occluded object recognition

Occlusion presents a significant challenge in the application of object recognition tasks, particularly within the complex environments of construction sites. The dynamic interplay of objects, equipment, personnel, and machinery in three-dimensional construction spaces frequently results in visual obstructions. These challenges are exacerbated by the inherently dynamic and cluttered nature of construction sites, which are characterized by variable lighting conditions, the presence of reflective materials, and the diverse colors and shapes of workers' attire. The issue is further complicated by various types of occlusions, such as self-occlusion, inter-object occlusion, and occlusion by background elements, all of which are common in the dense and congested workspaces typical of construction sites. Such occlusions compromise the effectiveness of deep neural network-based object recognition algorithms relative to human perception and adversely affect the performance of detection systems. Despite numerous datasets available for object detection, a critical issue remains the scarcity of annotated data for occluded objects within construction environments, lacking ground truth labels for such objects. This gap highlights the challenge that existing large-scale datasets do not adequately represent potential occlusions and deformations [186]. In response to these challenges, [187] proposed an innovative approach by integrating Deep Convolutional Neural Network (DCNN) with compositional models to develop a Compositional Convolutional Neural Network (CompositionalNet). This approach diverges from traditional DCNN frameworks by incorporating a differentiable generative compositional layer in place of the fully connected head, thereby enhancing the network's capability to classify and localize partially occluded objects effectively. Their findings demonstrate that this method significantly outperforms conventional DCNNs and other related approaches in handling partially occluded objects. Additionally, Generative Adversarial Network (GAN)-based de-occlusion methods have emerged as a promising area of research, as

evidenced by a recent study that developed a two-stage Occlusion-Aware technique capable of distinguishing between occlusions and de-occluded objects [188]. These recent advancements can be harnessed to formulate Deep Convolution Neural Networks specifically designed for recognizing safety hazards within construction environments.

5.2.3. Limited object visibility

The efficacy of real-time safety hazard detection in dynamic construction environments is contingent upon the clear visibility of the object of interest. Factors such as dust, camera shake, variable environmental conditions (e.g., haze and illumination), and camera hardware limitations significantly degrade the quality of images captured at indoor construction sites. The resultant degradation in image quality adversely affects the efficacy of image feature engineering, culminating in a diminished performance in the detection of fall hazards. In response to this challenge, [189] implemented the AtJ-DH network, an image dehazing and denoising model [190]. This model employs a shared encoder structure for the extraction of informative features from the contaminated images. Subsequently, this approach employs three dense decoders to integrate the denoised feature maps derived from the encoder, thus effectively reducing noise and generating high-quality processed images. The incorporation of such autoencoder-based dehazing and denoising methodologies into current Deep Convolutional Neural Networks (DCNNs) has the potential to substantially enhance visibility for the real-time detection of safety hazards, consequently reducing the occurrence of false-positive alarms.

5.2.4. Small object detection

Despite significant advancements in large and medium-sized object detection through deep learning methodologies, detecting small objects from construction scene is still a difficult and challenging problem in the field of computer vision due to the limited size, less appearance, and geometry cues, and the lack of large-scale datasets of small targets. Enhancing the efficacy of small object detection carries broad implications for real-world safety applications within the construction sector. For example, a vision-based safety monitoring system designed to detect small hazardous objects (e.g., safety lines, drills, screws, hammers, and nails) are required to be detected effectively to ensure the safety of the on-site construction workers. Extensive research and reviews have been done to address the issue previously. [191] reviewed the methods of detecting small objects from five aspects, including multi-scale feature learning, data augmentation, training strategy, context-based detection and GAN-based detection. Determining an appropriate detection module for safety applications presents a challenge due to the paucity of construction-specific data featuring small objects. Nonetheless, this study identifies the potential of Generative Adversarial Networks (GANs) to detect small objects in construction environments, attributed to their capability to enhance image quality from blurry inputs. [192] proposed a pragmatic approach proposing an end-to-end Multi-Task

Generative Adversarial Network (MTGAN). In the MTGAN, the generator functions as a super-resolution network capable of up-sampling small, blurred images into high-resolution counterparts, thereby recovering detailed information for enhanced detection accuracy. Concurrently, the discriminator operates as a multi-task network, evaluating each input image patch based on a real/fake score, object category scores, and bounding box regression offsets. To facilitate the generator in recovering more detailed information for simplified detection, the classification and regression losses identified by the discriminator are backpropagated into the generator throughout the training process.

5.3. Audio data-based automated safety

Table 8 highlights the most common challenges in applying audio-based safety automation and provides recommendations for future research. These are further discussed in the subsequent sections in detail.

5.3.1. Noise source identification

Construction sites are characterized by diverse types of sound sources, including equipment operation, construction activities, and worker communications, creating a complex acoustic environment. Identifying source noise from such mixed noisy environments as a part of automated site monitoring has been one of the biggest challenges in audio-based safety research. For example, distinguishing the warning beep of a reversing vehicle from the general background noise requires advanced classification techniques. Previous studies have demonstrated the effectiveness of Deep Convolutional Neural Networks (DCNN) in identifying both single and mixed noise sources, leveraging audio spectrograms to facilitate direct mapping for sound-based activity monitoring [169,193,194]. Recent advancements have focused on capturing long-range global context from audio spectrograms to improve the source identification accuracy by integrating attention mechanisms with DCNN [195]. These hybrid CNN-attention models have set benchmarks in various audio classification tasks, including audio event classification, speech command recognition, and emotion recognition. A novel approach challenged these methods by introducing Audio Spectrogram Transformer (AST), a convolution free model that applies purely attention-based mechanisms directly to audio spectrograms [196]. This method, drawing on the success of vision-based transformer models (ViT) in the vision domain has shown improved accuracy in automating noise identification and level estimation in mixed-noise construction environments [197]. Such advancements underscore the potential of AST in improving noise management within inherently noisy construction environments.

5.3.2. Quality data availability and annotation

The efficacy of developing a sound recognition system in classifying diverse range of sounds within construction sites largely depends on the quality and comprehensiveness of the training datasets. Given the vast range of potential noise sources on a construction site, collecting a sufficiently diverse and representative data set has significant challenges. This situation is compounded by the lack of specific safety-critical sound events, leading to data imbalance, and the difficulty of accurately annotating complex construction noises, which requires specialized knowledge. To address these issues, data augmentation and self-supervised strategies have been recommended to enrich training datasets, thereby improving the model's performance by enhancing the variety and balance of training samples.

Previous studies have leveraged deep neural network models for sound and audio classification tasks due to their over-parameterized nature, necessitating large datasets to address overfitting and reduce sensitivity to background noise and information redundancy [198,199]. Data augmentation techniques play a crucial role in environmental sound classification by generating artificial input data from existing

Table 8

Summary of the challenges and recommendations of audio-based safety applications.

Challenges	Recommendations
Noise Source Identification: Addressing the differentiation between hazardous noise from ambient site noise.	- Addressing the differentiation between hazardous noise from ambient site noise
Quality Data Availability and Annotation: Lack of annotated sound dataset diminishing the accuracy of the classifiers.	- GAN based data augmentation. - self-supervised audio-data annotation.
Feature Representation: Determining the best feature representation to improve classification accuracy.	- MFCC for better feature representation

sound samples, preserving essential information for the task while introducing variation. Advanced data augmentation strategies, such as the use of Generative Adversarial Networks (GANs), along with their variants including ACGAN, WCCGAN, combined GAN and VAE, and WaveGAN, have been identified as state-of-the-art in enhancing datasets for sound classification, speech recognition, and music retrieval [200]. Despite the advancements in data augmentation techniques, challenges such as potentially limiting the structural complexity of neural networks and generating datasets that may not accurately represent the intended task, leading to model overfitting [200]. Furthermore, the efficacy of sound recognition systems in noisy construction settings is hindered by a scarcity of annotated data, necessitating expert human intervention and considerable labor hours. To address this challenge, Gong et al. demonstrated a method to enhance audio and speech classification by reducing the reliance on extensive labeled datasets. The study introduced a self-supervised learning approach for the Audio Spectrogram Transformer (AST) model, utilizing unlabeled audio data through joint discriminative and generative masked spectrogram patch modeling (MSPM), offering a promising direction for future research in sound classification within noisy environments [201].

5.3.3. Feature representation

In addition to integrating an effective data augmentation technique, determining the best feature representation method from diverse range of sounds generated in construction sites play a vital role for improving sound classification accuracy. Recent research trends emphasize the development of classifiers based on Deep Neural Networks (DNN) to achieve robust accuracy. These models typically necessitate input data formatted as matrices and interpreted as images. Mel Frequency Cepstral Coefficients (MFCC) and Mel Spectrograms are the preferred feature extraction methods, enabling the transformation of one-dimensional audio sequences into image-like formats suitable for deep learning model training. MFCC has exhibited state-of-the-art performance in sound classification, being highlighted in approximately 26% of research studies [200]. The MFCC format processes audio samples through a series of steps, initially applying a Fourier transform to capture spectral powers. These powers are then mapped onto the mel-scale, with the logarithmic values of these mel-scale frequencies calculated. Subsequently, a discrete cosine transform is utilized on the sequence of mel-logarithmic powers, with the resulting MFCCs representing the spectrum's amplitudes. In contrast, the log-mel spectrum is commonly applied omitting the final transformation step. Therefore, deep learning-based features, while computationally intensive and data-demanding, can be optimally replaced by the compactness of semantic features for effective recognition of construction events among dynamic background noise.

6. Study contributions

This study contributes to the field of construction safety by providing a comprehensive review of AI applications, with a particular focus on text, visual, and audio data. It categorizes AI applications based on data modality, offering a clear framework for understanding the distinct characteristics of AI models and challenges associated with each type. This systematic categorization aids in delineating how various AI technologies can be effectively utilized in construction safety contexts. For researchers, the study highlights current trends, gaps in existing literature, and suggests potential directions for future research, especially in refining AI methodologies for construction safety applications. Practitioners in the construction industry can benefit from the study's findings by gaining a deeper understanding of how different types of AI applications can be leveraged to enhance safety on construction sites. The study provides a nuanced view of the strengths and limitations of AI applications using text, visual, and audio data, aiding practitioners in making informed decisions about implementing these technologies in their safety protocols. Additionally, the study's insights into the

challenges of AI integration in construction safety can inform the development of practical strategies for overcoming barriers to AI adoption in the industry.

7. Conclusion

In the global construction industry, high rates of work-related injuries and deaths highlight the urgent need for enhanced safety interventions. Artificial Intelligence (AI) is increasingly recognized as a significant tool for enhancing construction safety management. On construction sites, a wealth of data mainly in the form of images, videos, text, and audio is generated, offering a rich resource for AI applications. With advancements in computational hardware and algorithms, there is potential to harness this data to strengthen safety measures. This review presents the current state of AI usage in improving construction safety, identifies the challenges that impede its adoption, and provides recommendations for research opportunities, encouraging the development of different AI solutions for enhancing construction safety management. The review categorized the literature based on the primary datatype - text, visual, and audio data as these are the most prevalent forms of data used in AI applications within the construction safety domain.

The review indicated real-time safety monitoring, hazards detection, and information extraction are the most popular fields where the applications of AI are explicitly observed. Among different methods and techniques demonstrated in the study, object recognition and NLP methods are found prevalent with the growing interest in audio classification to enhance construction safety. Commonly, deep neural network models are being employed across AI-based safety management methods.

The review also identified different challenges in implementing AI-based safety systems in the complex construction environment. A key issue across all AI applications is the need for high-quality annotated data management and processing. Text applications additionally grapple with unstructured language and specialized terminology used in construction. The visual systems are limited by the intricacies of image processing especially occlusions, lighting variations, perspective and angle issues, and diverse construction objects and activities. On the other hand, audio systems face a challenge of discerning different construction sounds in mixed noisy environments. Integrating these AI methodologies with current workflows presents another challenge as reported by multiple publications. Text based systems need to work with existing documentation processes, visual systems require to tackle detecting small and occluded objects, and audio systems also need frequent adjustments due to environmental noise variations.

Future research should prioritize the development of integrated systems harnessing the synergy of text, visual, and audio data for multifaceted safety monitoring. While audio-based AI systems remain underexplored, they hold substantial promise for detecting auditory cues indicative of unsafe practices or hazards. With the advent of Large Language Models (LLMs), there is also a considerable opportunity to advance text-based safety applications by interpreting complex construction terms and documentation. Moreover, the exploration of AI in predictive analytics and automated safety regulation compliance represents a significant frontier. Ethical considerations, particularly in privacy and bias, are imperative in guiding the responsible integration of AI into construction safety practices. By focusing on these areas, research can facilitate safer construction sites, mitigating injury risks through innovative AI applications.

Credit authorship contribution statement

Ahmed Bin Kabir Rabbi: Writing – review & editing, Writing – original draft, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Idris Jeelani:** Writing – review & editing, Resources, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors affirm that they have no identifiable financial conflicts or personal affiliations that might have seemingly impacted the findings presented in this paper.

Data availability

Data will be made available on request.

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